Two Kinds of Saturation: Why LLM-Judge Order Bias Is Hard to Measure

privatedick*

August 2026

Abstract

Anyone who has used an LLM as a pairwise judge has watched it flip its verdict when the two candidates are swapped. The standard remedy—run both orders and compare—answers neither of the questions underneath: *why* does order matter, and why is *measuring* the effect so easy to get wrong? We separate two mechanisms that the word "saturation" conflates. *Internal* saturation (a clamped accumulator) can **create** order sensitivity: we formalize a clamped belief update and prove, in Lean 4, that constant-increment updates commute while the same update under a clamp does not. *Output* saturation (near-deterministic judge verdicts) does the opposite—it **destroys** the observability that order statistics require. Empirically, using a fully crossed 2×2 design that separates content-position effects from label-preference effects (a confound in our own first design, corrected here), we measure three open-weight judges and one proprietary frontier judge: 64–68% of judge conditions are saturated, 14 of 18 (model, item) cells fail an exact-test interpretability gate, effective sample sizes collapse to $\approx$1.1, and order effects are null after Holm correction (0/18, replicated across two independent runs). On the same tie items where order effects were near zero, label bias was 7–10$\times$ larger—item-specific, not a universal position prior. A statistical power analysis shows the exact saturation gate has *zero* power below k=29 samples per condition at the tested threshold, so quick-start sample sizes cannot in principle detect what they claim to measure. The methodological point: before asking how large an order effect is, ask whether the judge's output distribution contains any variance to order. We release the crossed design, the gates, the Lean proofs (0 sorry, axiom-audited at build time), and an offline proof-of-concept; total experimental cost was $0.46.

1 Introduction

LLM-as-judge evaluation asks a model "which of these two answers is better?" and records the verdict. The verdict is not stable under swapping the two candidates, and the field's standard fix is procedural: query both orders, check agreement, average. The fix works well enough in practice, but it leaves two questions unanswered. First, *why* does order matter at all—what is the mechanism? Second, why does *measuring* the effect keep going wrong in ways that the both-orders protocol does not catch?

This paper's claim is that the word "saturation" is doing double duty over two mechanisms with opposite consequences:

1. **Internal saturation can manufacture an order effect.** A bounded belief accumulator with a clamp—$x \mapsto \text{clamp}(x + \delta)$—has state-dependent increments, and state-dependent

*Pseudonymous author. All code, proofs, and raw results are public and reproducible at https://github.com/privatedick/llm-judge-saturation (MIT); nothing in this paper rests on trusting the author.

increments do not commute. We prove the positive and negative halves in Lean 4 (§3): clamped updates commute in the interior of the range and provably fail to commute at the boundary.

2. **Output saturation can hide one—or make its measurement meaningless.** A judge whose per-condition output distribution is near-deterministic still produces a point verdict, but no variance, no error bars, and effectively one observation. Order statistics computed on such a distribution are non-identifying. Kang [4] states this as a caveat for QQ-equality audits; we make it operational as an exact test that gates every downstream statistic (§5).

The empirical headline is a null result with a decomposition attached: after separating position from label preference by full counterbalancing, we find *no* Holm-significant order effects in any of 18 cells across two independent runs, on judges where 64–68% of conditions are too saturated for the question to be well-posed—while label bias on the same items is an order of magnitude larger and item-specific. Our own first design confounded the two signals and initially suggested an order effect; the crossed rerun killed it. We report that correction in place (§7) because the failure is the methodological point of the paper.

2 Background: order effects as non-commutation

Write T_A, T_B for the state updates of processing item A then B. An order effect is

$$T_B(T_A(x)) \neq T_A(T_B(x)),$$

i.e. the updates fail to commute. Quantum-cognition models instantiate this with vectors and projectors and derive a parameter-free structural invariant, the **QQ equality** [1, 2]:

$$p(A_y, B_n) + p(A_n, B_y) \;=\; p(B_y, A_n) + p(B_n, A_y),$$

both orders agree on total "changed my mind" probability, however large the individual order effect. Two caveats we carry throughout: satisfying QQ does not establish the Hilbert-space model [3], and QQ is one structural invariant among several, not a structure-vs-noise oracle.

3 Internal saturation creates non-commutation

Constant additive updates, $x \mapsto x+\delta$ for fixed δ, always commute—translations compose order-free. The cheapest way an *additive* update becomes order-sensitive is a clamp:

Definition 1 (Clamped update)**.** $\mathrm{clamp}_{[lo,hi]}(y) = \max(lo, \min(hi, y))$*, and the update* $U_\delta(x) = \mathrm{clamp}_{[lo,hi]}(x+\delta)$.

Both halves are machine-checked in Lean 4 (repository, `lean/EvoEcos/ClampedUpdateCommute.lean`, 0 sorry, with a build-time axiom audit that fails the build on any axiom outside Lean's standard three):

Lemma 1 (Interior commutation)**.** *If* x, $x+a$, $x+a+b$ *and their mirrored order all lie in the open interior of* $[lo, hi]$*, then* $U_b(U_a(x)) = U_a(U_b(x))$*: within the interior the clamped update is an affine translation and translations commute.*

Proposition 1 (Saturation breaks commutation)**.** *There exist lo, hi, x, a, b with the accumulator in range but at the boundary where $U_b(U_a(x)) \neq U_a(U_b(x))$: the effective increment becomes a function of state, and order sensitivity appears exactly where the accumulator saturates.*

The same repository proves the perturbation companion used for framing/order sensitivity bounds (`PersuasionOperator.lean`): the persuasion power satisfies $|\text{power}(\varphi, \theta)| \leq |\sin\theta|$ and the bound is tight. The two results together say: internal saturation is a *sufficient* mechanism for order sensitivity (not a claim that production LLMs implement clamped accumulators), and bounded perturbation is the right frame for asking how far order or framing can move a decision.

4 Output saturation destroys observability

Kang [4] audits question-order effects in LLMs with the QQ equality and flags the caveat that a near-deterministic judge makes the QQ comparison non-identifying: with no variance there is no "changed-my-mind" probability to compare across orders. We treat that caveat as a first-class measurement gate rather than a footnote.

Definition 2 (Saturation gate)**.** *A condition's verdict distribution is* saturated at level s *when the top outcome probability exceeds s under an exact one-sided binomial test ($H_0 : p_{top} \leq s$), not a bare point-estimate threshold: a point-estimate check over-flags dispersed judges from finite-sample noise.*

We report saturation at two levels because they answer different questions: the fraction of individual (position, label) conditions that are saturated, and the fraction of (model, item) cells in which *any* of the four conditions is saturated—the conservative gate under which order statistics for that cell are non-identifying.

The gate has a power floor that quick-start designs silently violate. At $s = 0.9$ and $\alpha = 0.05$, the exact test has *zero* power below $k = 29$ samples per condition: at $k \leq 28$ even an all-agree verdict string ("AAAA…") cannot reject the null of non-saturation. The crossover at $k = 29$ is verified in the repository's test suite by running the real exact test on synthetic all-agree lists at k=28 and k=29 (`python/test_gate_power.py`). Our first-pilot defaults sat below the floor; the repository's earlier quick-start did too.

5 Design: crossing position against label

A judge asked "A or B?" can prefer the first slot (a *position* effect) or prefer the token "B" (a *label* effect). Every A/B-then-B/A protocol without label counterbalancing measures their sum. The design used here is a fully crossed 2×2 per item: (content order) $\times$ (label assignment), $k = 40$ samples per condition, temperature 1.0. Decomposition:

- **order effect** = the pure content-position signal (same labels, swapped content), Fisher exact test per cell, Holm correction across cells;
- **label bias** = the pure token-prior signal (same content, swapped labels).

Three open-weight judges via OpenRouter (DeepSeek v4-flash, Qwen3-8b, Mistral NeMo) on six items (three "tie" items with no clear winner, three clear-winner controls), giving 18 (model, item) cells and 72 conditions per run; two independent runs on different days. A proprietary frontier judge (GPT-5-mini) was added later on a three-item subset. Total API cost across all three runs: $0.46.

6 Results

	Run 1 (08-11)	Run 2 (08-12)
conditions saturated (s=0.9, exact test)	46/72 (63.9%)	49/72 (68.1%)
cells failing interpretability gate	14/18	14/18
interpretable cells	4	4
mean order effect, all cells	0.018	0.024
mean order effect, interpretable	0.041	0.063
Holm-significant order effects	0/18	0/18
mean label bias, tie items	0.456	0.408
mean order effect, same tie items	0.046	0.058
mean effective sample size (worst condition)	1.10	1.10
estimated cost	\$0.21	\$0.21

Table 1: Two independent runs, three open-weight judges. The null replicates: no cell shows an order effect surviving Holm correction; on the tie items, label bias is 7–10× the order effect measured on the same items.

Three findings, in increasing order of interest:

(1) The order-effect null replicates. Zero of 18 cells per run, twice. One cell in Run 2 reached raw $p < .05$ and did not survive Holm; at this k, a difference of one sample is not a measured magnitude.

(2) Most judge conditions are too saturated for the question to be well-posed. Between 64 and 68% of conditions and 14/18 cells fail the exact gate; effective sample size is ≈1.1. The judge returns a verdict, but the verdict stream carries almost no information about *ordering* of anything.

(3) Label bias dwarfs order bias and is item-specific. On the tie items—precisely where a position prior would express itself—mean label bias was 0.41–0.46 against order effects of 0.046–0.058 on the same items. The frontier judge sharpened the picture: GPT-5-mini showed label bias of 0.49 on one tie item (essentially unbiased) and 0.11 on another (strongly B-preferring)—the same judge, two items, opposite characters. That is evidence for item-specific token quirks, not a universal position prior.

7 The correction, reported in place

This paper's first design was wrong, and we report the failure because it is the methodological point. The original A/B-then-B/A protocol confounded content position with label preference; it initially suggested an order effect. The crossed rerun decomposed the signal, and the effect moved entirely into the label channel. Every number above is from the crossed design.

The correction went both directions. An external review (2026-08-12) claimed four reproductions of a parser bug and one missing length-check; when actually run, three of the four parser examples returned the correct letter against the code as shipped, and the length-check was already present and already raising. The fixes that were real were applied only after reproducing the failure directly—the same discipline this paper argues for, applied to its own reviewers. Two review-requested extensions are also reported honestly: the proprietary-judge extension succeeded (above); a temperature sweep ($T \in \{0, 0.5, 1\}$) was attempted twice, failed twice on the same infrastructure cause (a per-read inactivity timeout that a trickling connection never trips; now a hard

wall-clock deadline), and is *not* reported from incomplete runs. The temperature question stays open.

8 Limitations

The judges are three open-weight models plus one frontier model on a six-item panel: this is a measurement-methodology demonstration, not a benchmark. The Lean results establish clamped saturation as a *sufficient* mechanism for non-commutation, not a claim about transformer internals. $k = 40$ per condition is above the gate's power floor but modest for small effects. The QQ equality is used as a structural reference point, and its satisfaction would not establish the Hilbert-space model [3].

9 Reproducibility

Everything is public at https://github.com/privatedick/llm-judge-saturation (MIT): the crossed experiment and its tests, the exact gates with the k=29 power-floor test, the Lean 4 proofs with the build-time axiom audit, raw result JSONs for all runs, and an offline proof-of-concept requiring no API key. CI runs the full suite on every push; the badges in the README are enforced, not asserted. Total experimental cost: $0.46.

References

[1] Z. Wang and J. R. Busemeyer. A Quantum Question Order Model Supported by Empirical Tests of Order Effects on Survey Measurements. *Topics in Cognitive Science* 5(4):689–710, 2013.

[2] Z. Wang, T. Solloway, R. M. Shiffrin, and J. R. Busemeyer. Context effects produced by question orders reveal quantum nature of human judgments. *PNAS* 111(26):9431–9436, 2014.

[3] M. Boyer-Kassem, S. Duchêne, and E. Guerci. Testing quantum-like models of judgment for question order effect. arXiv:1501.04901, 2016.

[4] P. Kang. Auditing Question-Order Effects in Large Language Models with the QQ Equality: Mechanism Characterization and a Saturation Caveat. arXiv:2607.17219, 2026.